

XAI for Forest Fire Risk Assessment

Visualizing Meteorological Drivers with SHAP
Frameworks

University of Trieste | Data Visualization
June 9th, 2026

01. RESEARCH CONTEXT

Dataset Summary

Origin: Montesinho Natural Park (Portugal).

Telemetry: Real-time weather data (Temp, RH, Wind, Rain).

Objective: Predict burned area using Random Forest Regressors.

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Analytical Challenge

Machine Learning models are typically "black-boxes."

By applying **SHAP (SHapley Additive exPlanations)**, we map mathematical weights into intuitive **Visual Channels** for human interpretation.

01.1 DATASET AT A GLANCE

Variable Overview (13 total)

Variable	Type	Description
X, Y	Spatial	Coordinates in Montesinho Park
month, day	Temporal	Date of the observation
FFMC, DMC, DC, ISI	FWI Indices	Derived fire risk indices (not used)
temp, RH, wind, rain	Meteorological	Used in model (directly observable)
area	Target (ha)	Burned area to predict

Focus: 4 meteorological variables only, isolating direct atmospheric conditions from derived FWI indices.

Key Statistics

517

total records

47.8%

fires with area > 0 ha

69%

records in Aug-Sep

1,091 ha

largest fire recorded

Meteorological Ranges

Temp:

2.2 - 33.3 °C (mean 18.9°C)

RH:

15 - 100 % (mean 44.3%)

Wind:

0.4 - 9.4 km/h (mean 4.0)

02. ADDRESSED QUESTIONS

1. How do individual meteorological factors globally impact the prediction of fire risk?

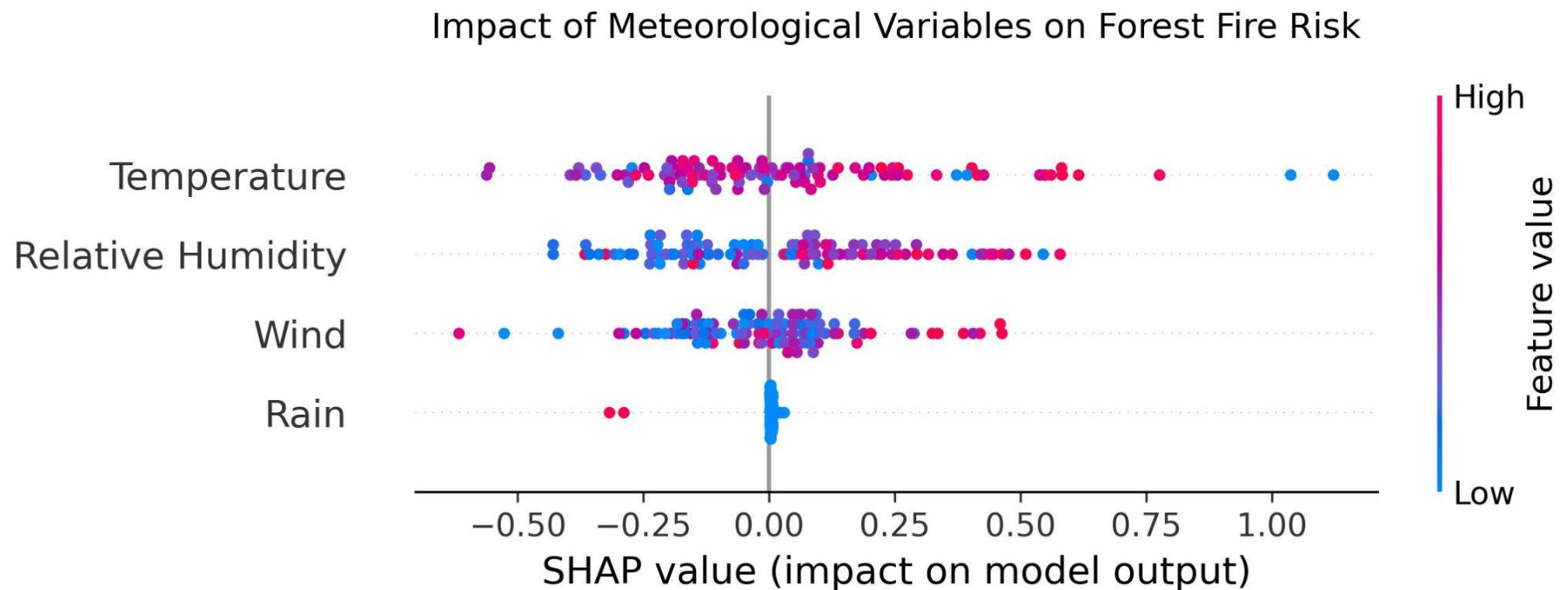
02. ADDRESSED QUESTIONS

1. How do individual meteorological factors globally impact the prediction of fire risk?

2. What specific weather triggers drove the prediction for a high-risk case study (Day 5)?

03. GLOBAL IMPACT ANALYSIS

The model captures counterintuitive relationships: higher temperatures unexpectedly lower predicted risk, exposing complex variable interactions.



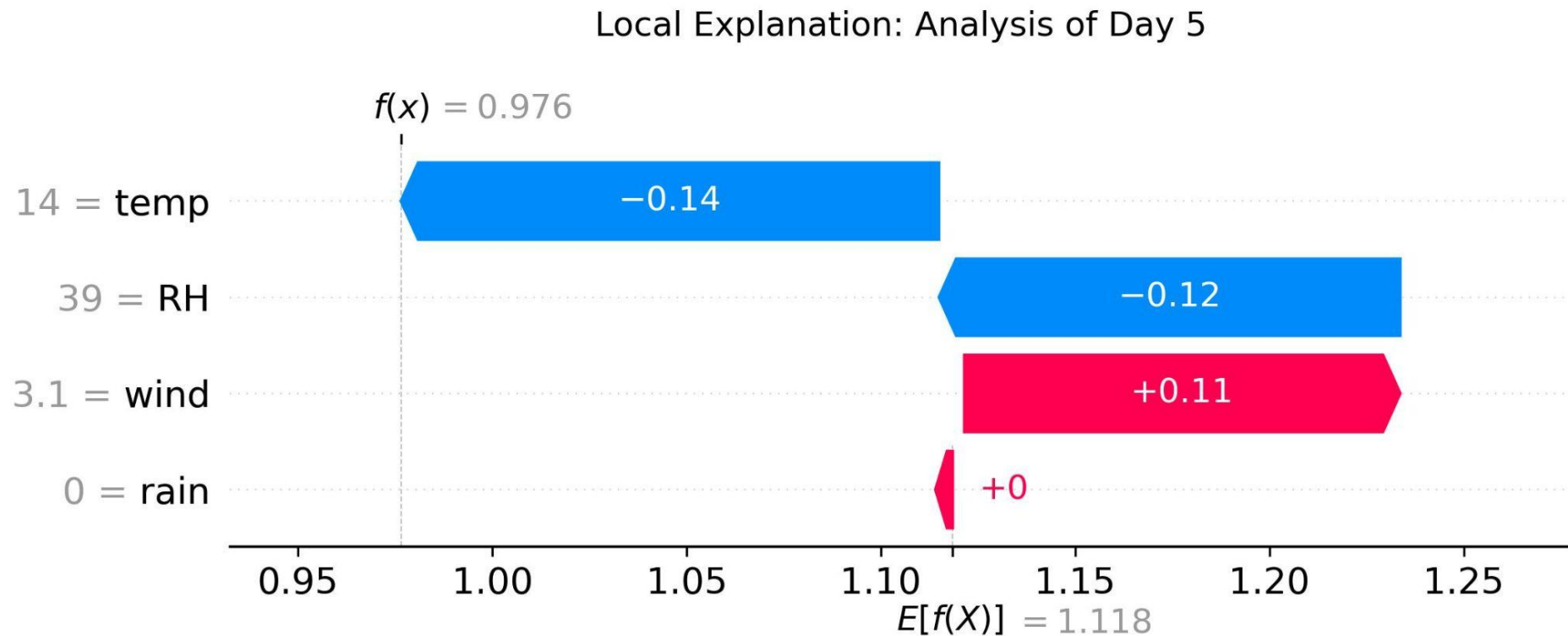
03. MAPPING: GLOBAL VISUALIZATION

Data Attribute	Attribute Type	Visual Channel Assigned
Feature Importance	Nominal / Ranked	Vertical Position (Y-axis Order)
Impact on Prediction	Quantitative	Horizontal Position (X-axis)
Predictor Value	Quantitative	Color (Diverging Blue-to-Red)
Sample Density	Quantitative	Vertical Jitter (Point Clusters)

Note the **non-linear interactions**: even at low temperatures (blue dots), extreme wind or drought conditions can push risk to the right.

04. LOCAL EVENT BREAKDOWN

On Day 5, the cooling effects of temperature (14) and humidity (39) outweighed wind impact (+0.11), dropping the final risk prediction to 0.976.



04. MAPPING: LOCAL VISUALIZATION

Data Attribute	Attribute Type	Visual Channel Assigned
Expected Baseline	Quantitative	X-Axis Origin Point
Feature Contribution	Quantitative	Bar Horizontal Width (Size)
Impact Direction	Categorical	Color (Pink: Increase / Blue: Decrease)
Final Output Area	Quantitative	Final X-Axis Endpoint

This sequential decomposition (Waterfall) minimizes cognitive load while identifying specific environmental triggers for a single prediction.

05. INDIVIDUAL CONTRIBUTIONS

Student Contributions

Arahí Fernández Monagas

- Dataset selection and exploration
- Random Forest model training and evaluation
- SHAP explainability framework (global + local)
- SHAP Beeswarm and Waterfall visualizations
- Presentation design and narrative

Tools: Python (sklearn, shap, matplotlib, pandas) • PowerPoint

★ Bonus: Discarded Visualization

Discarded: SHAP Bar Chart (mean |SHAP|)

A simple bar chart ranking features by average absolute SHAP value was considered as an alternative global explanation.

Why rejected: collapses distribution to a single mean, losing direction of impact and non-linear behavior. The beeswarm encodes more information per visual channel with no added cognitive load.

Beeswarm advantage: shows magnitude, direction AND feature value (color) simultaneously for all samples, revealing non-linear interactions that bar charts hide.

Conclusion

Black-box models can inherit historical data biases, learning **false physical relationships**. **XAI** visualizations act as a critical audit tool, exposing these **flaws to prevent blind trust** in automated systems
